

Academic Review of *A hypothesis of ape and human evolution based on comparative biology and plate tectonics* (Verhaegen et al., 2026)

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1 General Positioning of the Paper

This article belongs to the long tradition of semi-aquatic hypotheses of human evolution, from Hardy and Morgan to more technical modern variants such as aquarborealism, littoral foraging, and wading-first bipedalism. The authors propose a global alternative model that reinterprets hominoid evolution, questions the hominin status of australopiths, suggests an extra-African origin for *Homo*, and integrates plate tectonics into the evolutionary narrative. The paper is synthetic, programmatic, and non-empirical.

2 Analytical Summary of the Content

2.1 Miocene Aquarboreal Model

The authors argue that Miocene hominoids lived in flooded forests, combining bipedal wading, vertical climbing, and below-branch locomotion. They write that these apes “engaged in frequent bipedal wading and below-branch vertical climbing” (p. 1). Tail loss, pelvic morphology, scapular position, and spinal structure are interpreted as adaptations to this lifestyle.

2.2 Australopiths as Non-Hominins

The paper claims that australopiths resemble *Pan* or *Gorilla* more than *Homo*, and that their “human-like” traits are ancestral rather than derived. The authors propose that *Homo* originated outside Africa and later returned. They cite retroviral infection patterns (PtERV1/PtERV2) as

supporting geographic separation. For example: “*Ardipithecus ramidus* (~4.4 Ma) had small canines... while some later species... had larger ones” (p. 2).

2.3 Littoral Phase of *Homo*

The authors invoke pachyosteosclerosis in *H. erectus*, auditory exostoses, associations of fossils with molluscs, human subcutaneous fat, respiratory and oral morphology, and the multipapillary kidney. They state: “Very dense bones... are exclusively found in slow, shallow-bottom diving tetrapods” (p. 3).

3 Strengths of the Paper

3.1 A Useful Critique of the Savanna Model

The article highlights the historical and conceptual weaknesses of the savanna-to-bipedalism narrative. The authors write: “almost every aspect of the savanna hypotheses is contradicted by recent insights” (p. 1).

3.2 Ambitious Interdisciplinary Synthesis

The paper draws on comparative anatomy, paleoecology, taphonomy, physiology, fossil data, and tectonics. The coherence is debatable, but the integrative effort is genuine.

3.3 Clear and Accessible Writing

The text is well structured, readable, and extensively referenced.

4 Major Methodological Weaknesses

4.1 Primacy of Morphological Analogies

The central methodological issue is the reliance on resemblance-based reasoning. Examples include: dense bone implying diving, subcutaneous fat implying aquatic adaptation, the philtrum implying nasal closure underwater, and sinuses implying buoyancy. These are correlations, not demonstrations.

4.2 Lack of Falsifiable Predictions

The model offers no discriminating tests. It explains known traits retrospectively but predicts nothing.

4.3 Ad Hoc Hypotheses to Address Genetic Contradictions

The authors write: “We consider our hypothesis to be more parsimonious, suggesting that our ancestors were not in Africa at all...” (p. 2). This is an ad hoc move to preserve the model rather than a deduction from data.

4.4 Selective Use of Fossil Evidence

The article minimizes australopith fossils, overlooks robust data on *Ardipithecus*, overinterprets associations between *Homo* and molluscs, and does not address counterexamples. The selection is not neutral.

4.5 Confusion Between Compatibility and Causality

The paper conflates traits compatible with aquatic behavior and traits necessarily resulting from aquatic behavior. This is the core conceptual weakness.

5 Scientific Positioning

5.1 What the Paper Is Not

The article is not mainstream, not based on new empirical data, not falsifiable, and not compatible with modern genetic evidence.

5.2 What the Paper Is

It is an alternative hypothesis, a speculative but ambitious synthesis, a useful critique of certain dogmas, and a methodologically fragile contribution. It is marginal but not pseudoscientific.

6 General Conclusion

The paper is stimulating and creative. It challenges entrenched assumptions and revisits long-standing debates. However, its reliance on analogies, correlations, narrative scenarios, and ad hoc reasoning places it outside the domain of empirical evolutionary science. It should be read as a speculative proposal rather than a robust explanatory model.

7 Verdict

An ambitious alternative hypothesis, but methodologically insufficient. Interesting for reflection, not convincing as a scientific model.

References

Verhaegen, M., Munro, S., Bonne, K., Mansfield, F., Vaneechoutte, M. (2026). *A hypothesis of ape and human evolution based on comparative biology and plate tectonics*. Academia Biology, 4(2). <https://doi.org/10.20935/AcadBiol8310>